TechPad Bill of Materials

Kit parts only. No extra chips, no expander. You have spare switches, keycaps, diodes and LEDs.

Ref	Part	Qty	Where	Source
U1	Seeed XIAO RP2040 (through-hole mount)	1	Back of PCB	Hackpad kit
SW1 to SW12	MX-style mechanical switch	12	Top plate + PCB	Hackpad kit
-	White blank DSA keycap	12	On the switches	Hackpad kit
SW13	EC11E rotary encoder with push switch	1	Top right	Hackpad kit
D1 to D13	1N4148 diode, DO-35 through-hole	13	Next to each switch	Hackpad kit
LED1 to LED12	SK6812 MINI-E RGB LED (reverse mount)	12	Under each key	Hackpad kit
J1 + display	0.91" I²C OLED module, 4-pin header	1	Top left	Hackpad kit
-	M3 × 16 mm screw	4	From underneath, through PCB corners	Hackpad kit
-	M3 × 5 × 4 mm heat-set insert	4	Underside of the top frame, at the corners	Hackpad kit
-	TechPad PCB, 92 × 87 mm, 2-layer	1	-	PCB fab (from gerbers.zip)
-	Top frame, switch plate, bottom tray, knob cap	1 set	-	3D printed

LED power check **Likely**: 12 LEDs × about 60 mA at full white is about 0.72 A. Firmware caps brightness at about 40%, which keeps it near 0.3 A, comfortably within USB's 0.5 A.